

Robust and Accurate Multi-UAV Cooperative Semantic SLAM Leveraging Lightweight YOLO

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Abstract—Multi-UAV visual cooperative SLAM primarily uses cameras as sensors that work together to complete positioning and map generation. However, traditional SLAM systems are based on the assumption of a static environment and face problems of high computational load and poor robustness in dynamic environments. To address these issues, we propose a multi-UAV cooperative semantic SLAM system based on a lightweight You Only Look Once (YOLO) algorithm. First, we made lightweight improvements to the YOLO network with the aim of reducing the system's computational resource consumption and generating semantic information in real time. Second, we proposed an improved feature point screening algorithm that uses object detection and geometric constraints. This algorithm integrates semantic information and geometric constraints to judge the motion state of the target, excluding the feature points of moving targets, thereby improving the robustness and positioning accuracy of the system and making it adaptable to dynamic environments. Finally, we tested and evaluated our approach on the KITTI, TUM, and EuRoC datasets. The experimental results show that our system outperforms current mainstream SLAM systems in terms of feature extraction and positioning accuracy, and has good robustness and real-time performance.

Keywords—multi-UAV, collaborative SLAM, dynamic environment, feature extraction, feature selection

I. INTRODUCTION

Multi-UAV cooperative SLAM (Simultaneous Localisation And Mapping) technology is a key research area in autonomous UAV navigation[1]. Compared with the limited observation range, computational power, low efficiency and weak anti-interference capability of a single UAV, cooperative SLAM can consolidate data from multiple UAVs and effectively improve system accuracy, efficiency and robustness through information sharing and task distribution[2]. In recent years, cooperative SLAM systems have found wide applications in areas such as disaster relief, agriculture, geographic surveying and virtual reality. However, the increasing complexity of unknown environments and task requirements has raised the bar for multi-UAV cooperative SLAM technology.

Cooperative SLAM requires an effective system architecture to ensure data consistency throughout the system and to be responsible for data sharing and resource allocation. Schmuck et al. proposed the COVINS[3] cooperative SLAM framework, in which each front-end retains only basic visual odometry and shares local maps with the server. Liu et al.'s

RAMA-SLAM[4] takes resource competition into account, and by decoupling the server-side loop closure detection and the client-side VIO module, it effectively reduces the computational load and complexity of the cooperative localisation algorithm. However, these cooperative SLAM works overlook dynamic obstacles in real-world scenarios. In reality, there are often many moving objects in a scene. Traditional visual SLAM systems typically extract invariant features from the scene, such as corners and edges, and use them to infer changes in the camera's own pose. If there are moving objects in the scene, the feature points are likely to be distributed across the moving objects and change as these objects move. At this point, it is difficult for the camera to determine whether it is the scene or its own pose that has changed. Bescos et al.'s single-drone DynaSLAM[5] uses a detection method based on the Mask R-CNN[6] model to improve the accuracy of the system. However, the model they use is based on pixel segmentation, resulting in lower system efficiency. On the other hand, Wu[7] et al. use a more efficient YOLO[8] object detection model, which to some extent compensates for the lack of efficiency, but it cannot handle potentially moving objects. Therefore, in view of the lack of environmental responsiveness and autonomous navigation capabilities of multiple UAVs in dynamic environments, this paper presents a multi-UAV visually cooperative semantic SLAM system oriented towards dynamic environments.

Therefore, in response to the inadequacy of environmental adaptability and autonomous navigation capabilities of multiple UAVs in dynamic environments, this paper proposes a multi-UAV visual cooperative semantic SLAM system tailored for dynamic environments.

II. SYSTEM MODULES

A. System Overview

The overall framework of the multi-UAV cooperative semantic SLAM system based on lightweight YOLO proposed in this paper is shown in Figure 1. It mainly consists of three modules: a YOLO-based VIO front-end, a communication transmission module, and a server back-end. The VIO module ensures the autonomy of the UAV, and it mainly consists of four parallel modules: an object detection module, a dynamic feature selection module, a tracking module, and a local mapping module. The object detection module uses a lightweight YOLO V5s model and is responsible for detecting objects to provide semantic information to the UAV. The dynamic feature selection

module filters dynamic feature points based on the detected objects and the epipolar geometric relationship. The tracking module uses the resulting static feature points to complete keyframe-based tracking through feature matching and multi-view geometry, among other methods. The local mapping

module uses the keyframe information obtained from the tracking thread to further enrich the 3D map points and combines the minimisation of the reprojection error function and local bundle adjustment among other optimisation methods to construct a local point cloud map.

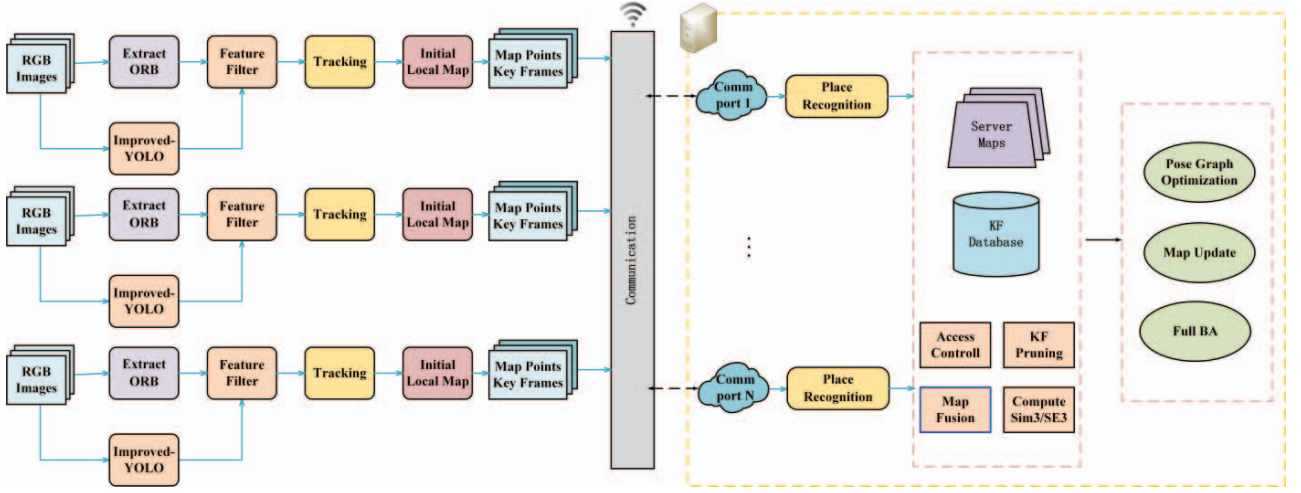


Fig. 1. System overview of the proposed multi-UAV SLAM.

To achieve efficient data exchange between the UAV and the server, the communication module adopts the message passing scheme proposed in the literature[9]. The UAV front-end only needs to transmit the static part of the feature points and keyframe information to the back-end. This approach can effectively reduce the amount of data transmission and improve the communication efficiency.

The server backend includes modules such as pose re-identification, loop closure and map fusion, and pose graph optimisation. The server first creates a keyframe database for loop closure detection. If it detects an overlap between the current keyframe and historical keyframes, it determines whether they belong to the same local map. If they belong to the same local map, the server runs the loop closure module, adds new nodes and edges for graph optimisation, and runs the non-linear optimisation module. If they belong to different local maps, the server runs the map fusion module and combines the bundle adjustment optimisation to construct a

large, highly accurate global map. In addition, to ensure the stability and robustness of the system, the server periodically performs operations such as deleting redundant frames, pose graph optimisation, and global bundle adjustment optimisation.

B. Enhanced Lightweight YOLO Object Detection Network

In UAV visual scenes, there are often many moving objects. If the feature points extracted by the algorithm are concentrated on dynamic objects, it will be difficult for the system to determine whether the change is in the scene or in its own pose, severely affecting the localisation and mapping system. To overcome this challenge, we use the YOLO algorithm to provide semantic information to the UAV, effectively eliminating the interference of dynamic objects. Furthermore, given the computational limitations of the UAV and the need for real-time operation, we have optimised the YOLO v5s object detection model for lightweight operation.

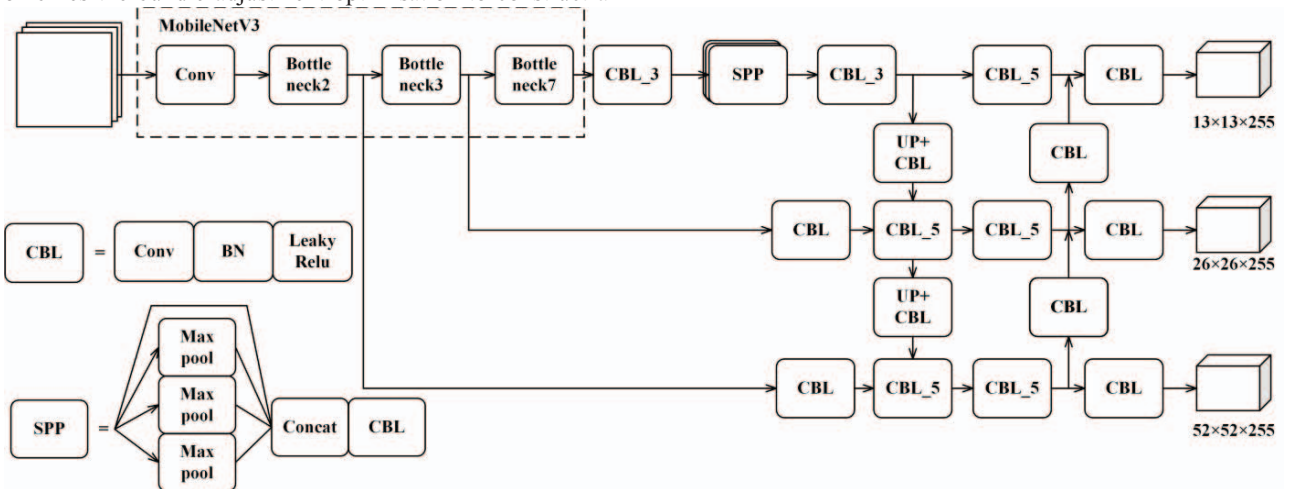


Fig. 2. Improved Lightweight YOLO Model Network Structure.

The improved lightweight YOLO network structure proposed in this paper is shown in Figure 2, and it mainly consists of four parts: Input, Backbone, Neck and Prediction. In the input part, we resize the original $640 \times 640 \times 3$ input image to $416 \times 416 \times 3$ to reduce the computational complexity and increase the processing speed. In the Backbone part, we replace CSPDarknet with MobileNetV3 for feature extraction, and use depth-separable convolutions instead of traditional convolutions, which reduces the number of model parameters and computational cost. In the Neck part, we integrate feature maps from different levels through a top-down pathway and lateral connections, based on the concept of Feature Pyramid Network. This captures multi-scale object information and improves detection accuracy and robustness. In the prediction part, we use a multi-scale detection strategy and generate feature maps of three sizes: 13×13 , 26×26 and 52×52 . We segment the image into grids corresponding to its size, create several prior boxes in the centre of each grid, and predict

C. Enhanced Algorithm for Motion Feature Point Selection

Object detection methods based on deep learning can provide object class labels, but they cannot accurately determine the current motion state of target feature points. For example, if the target is a human, it can be assumed that all feature points within the target box are motion feature points. However, if the target is a sedan, this cannot be directly excluded. Traditional multi-view geometry methods can eliminate outlier motion points, but when there are a large number of outliers in the scene, the basic matrix solution used to eliminate outliers has a large error, making it difficult for the algorithm to converge.

In response to the above problems, we propose an improved motion feature point selection algorithm that combines the YOLO algorithm with multi-view epipolar geometry constraints. This algorithm can effectively identify and track target objects in UAV scenarios, thereby improving the efficiency of UAV positioning and mapping. The principle of the algorithm is as follows.

The spatial relationship of the epipolar geometry is shown in Figure 3. Let the projections of a 3D point P in two images I_1 and I_2 be denoted p_1 and p_2 , respectively. The plane formed by the 3D point P and the camera centre O_1, O_2 is defined as the epipolar plane π . Consequently, the epipolar constraint equation is established.

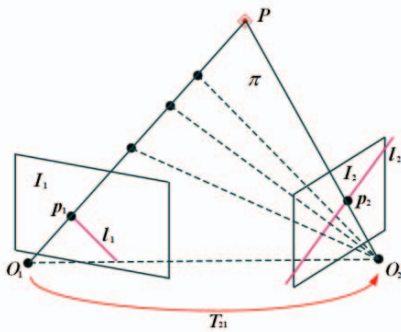


Fig. 3. Schematic Diagram of Epipolar Geometry Spatial Relationship

$$x_2^T E x_1 = p_2^T F p_1 = 0 \quad (1)$$

In the equation, x_1, x_2 represent the normalised coordinates on the pixel plane, E is the essential matrix and

whether the box contains an object and the object's category. Compared to the classic YOLO v5s model, our model is more efficient at handling objects of different scales and shapes, making it more suitable for scenarios where the UAV's computational capabilities are limited.

In a static environment, the epipolar constraint equation (1) can effectively describe the constraint relationship between spatial points and pixel points. However, in a dynamic environment, the spatial points and their corresponding pixel points often do not satisfy the epipolar constraint relationship. As shown in Figure 4, the pixel points of the spatial points in a dynamic environment will deviate from the epipolar line, and the 3D spatial points obtained by back-projection will have a large error from the actual spatial position. This error can cause the pose estimation based on the fundamental matrix to be inaccurate, thus affecting the positioning accuracy.

F is the fundamental matrix. Both can be converted by the camera internal parameter K .

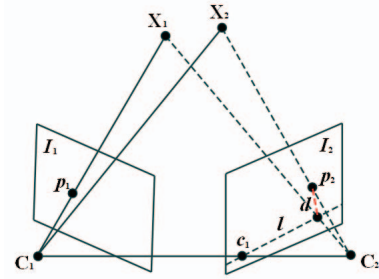


Fig. 4. Schematic Diagram of Geometric Constraint Relationships in Dynamic Environments.

We propose an improved motion feature point selection algorithm that eliminates motion feature points by calculating the distance from the feature points to the corresponding epipolar line. The specific approach is as follows: In the spatial coordinate systems of the left and right cameras, let the 3D coordinates of the spatial points be denoted as X_1, X_2 , and the corresponding 2D plane feature point coordinates as $p_1 = [u_1, v_1, 1]^T$ and $p_2 = [u_2, v_2, 1]^T$. Then the epipolar line l of the current frame can be determined through point p_1 :

$$l = F p_1 = F \begin{bmatrix} u_1 \\ v_1 \\ 1 \end{bmatrix} \quad (2)$$

Based on the formula for calculating the distance from a point to a line, the distance d from point p_2 to the epipolar line l can be determined as:

$$d = \frac{|P_2^T F P_1|}{\sqrt{\|A\|^2 + \|B\|^2}} \quad (3)$$

We consider the distance deviation of dynamic spatial points in two frames from their corresponding epipolar lines to determine whether the spatial point is in motion. Assume that the distances from the pixel points to the corresponding epipolar lines in the left and right views are d and d' respectively. The integrated distance $D = d + d'$ is defined and τ is the maximum distance for static points. If $D > \tau$, the spatial point is moving, otherwise it is stationary.

Compared to traditional SLAM systems, our proposed selection mechanism, which combines the YOLO object detection method and the epipolar geometry method, can reduce the disturbance of dynamic objects in unknown environments, thereby improving the accuracy and robustness of the visual front-end.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experiment Settings

To validate the effectiveness of the system proposed in this paper, we performed a performance evaluation of the algorithm based on publicly available datasets: TUM[10], KITTI[11] and the EuRoc[12] UAV dataset. The experiments were performed from three perspectives: the effect of UAV feature extraction, both quantitative and qualitative performance evaluation of the collaborative SLAM system, and system communication data. In addition, the system proposed in this paper was compared with the current mainstream SLAM systems to demonstrate its effectiveness.

The experimental platform used in this paper consists of server backend: desktop computer (Core i5-9600F @ 3.70 GHz*6, 16GB RAM); UAV frontend: Lenovo R720 laptop

(Core i5-7300HQ @ 2.50 GHz*4, 16GB RAM); communication intermediary: router (ZTE AX3000).

B. Dynamic Feature Point Selection

We first validate the effectiveness of the dynamic object elimination algorithm. Figure 5 shows the effect of feature extraction in four different environments. The first row shows common indoor and outdoor scenes, including familiar objects such as cars and people, with cars in both moving and stationary states. The second row shows the feature extraction results of the ORB-SLAM3[13], and the third row shows the feature extraction results of our multi-UAV collaborative semantic SLAM system based on lightweight YOLO. As can be seen, ORB-SLAM3 does not distinguish the motion state of objects and retains feature points on moving people and cars. These are used for localisation and mapping and thus affect the performance of the system. Our system detects objects such as vehicles in the environment using YOLOv5s, retaining feature points on stationary vehicles based on geometric relationships and eliminating feature points on moving vehicles. This avoids the influence of outlier features on localisation and mapping. The comparison validates the effectiveness of the dynamic object elimination algorithm proposed in this paper.

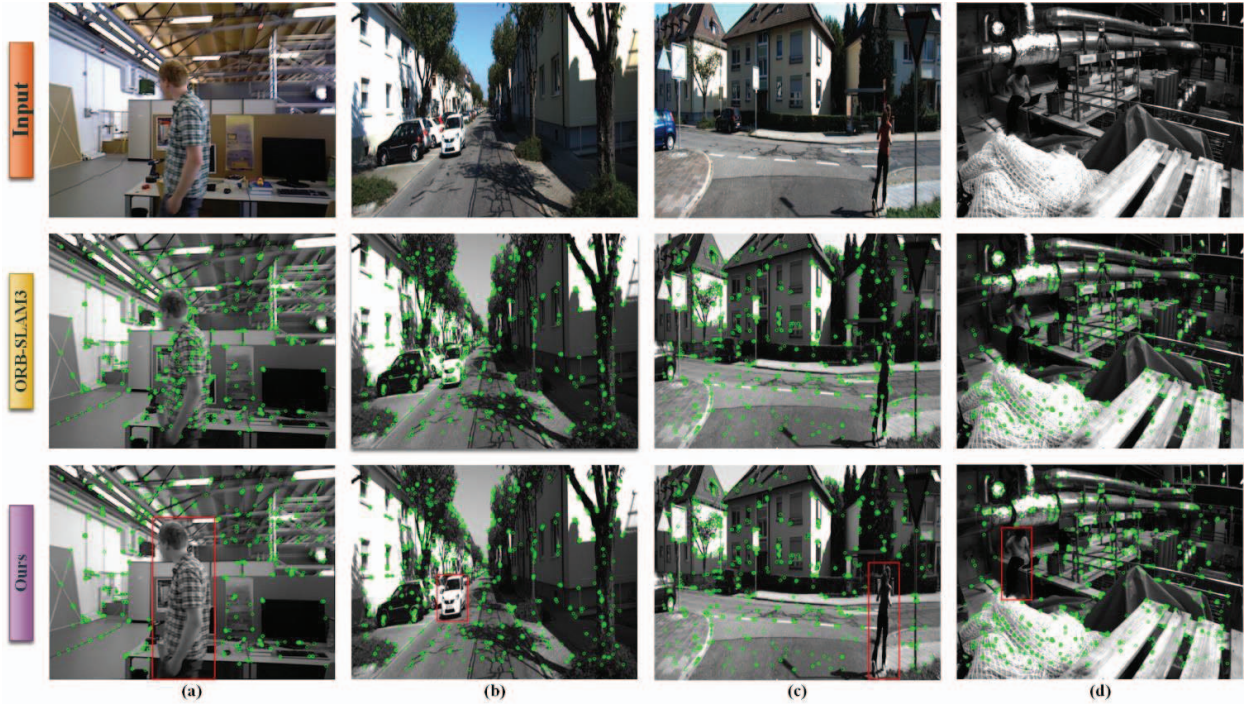


Fig. 5. Feature Extraction Performance of ORB-SLAM3 and Our System in Dynamic Environments.

C. Performance Evaluation of Multi-UAV SLAM

To evaluate the overall accuracy and robustness of the collaborative SLAM system, we compared our system with the COVINS system. Figure 6 shows the trajectory of our system. UAV 1 executes the first part of the KITTI 00 sequence, UAV 2 executes the second part of the KITTI 00 sequence, and UAV 1 executes the third part of the KITTI 00 sequence. The backend server is responsible for integrating the local maps, where the more red the colour, the greater the error, and the more purple the colour, the smaller the error. As can be seen, the backend server has successfully implemented loop closure detection and map fusion, and the accuracy of the global map is superior to that of the local maps.

The quantitative results of the comparison between our system and mainstream SLAM systems on the EuRoc dataset are shown in Table 1. The UAV front-end is responsible for executing the MH01, MH02 and MH03 sequences, while the back-end is responsible for constructing the global map. The bold black numbers indicate superior performance metrics. It can be observed that the improvements to the front-end of our system have reduced the trajectory error for the MH01 and MH02 sequences. However, there is less dynamic interference in the MH03 sequence, and the object detection thread wastes some resources, leading to an increased error in the MH03 sequence. Our server effectively utilises keyframes and map points in the local map to achieve the smallest trajectory

estimation error and optimal overall performance. In addition, VINS-mono[14] experienced initialisation or tracking failures in several experiments, resulting in an inability to build a global map, whereas our system ran normally several times. This further demonstrates the robustness of our system.

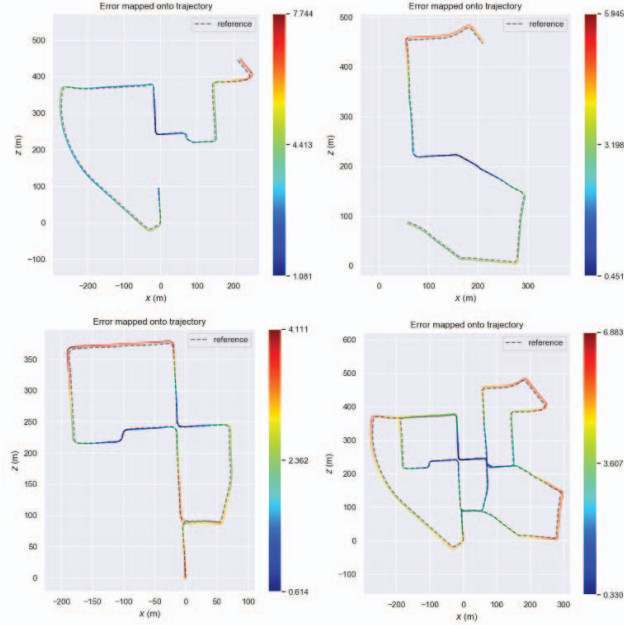


Fig. 6. Collaborative SLAM estimate with three agents using KITTI.

TABLE I. CAMERA LOCALIZATION RMSE ERROR (M) COMPARISON ON EUROC DATASET

System	MH01	MH02	MH03	MH01-03
VINS-mono	0.12	0.12	0.13	0.074
ORB-SLAM3	0.062	0.037	0.046	0.037
COVINS	0.052	0.041	0.069	0.026
Ours	0.046	0.033	0.054	0.021

D. Communication Statistics

Table 2 shows the network traffic generated between the server and the UAVs in our experiment. The UAVs send local map information to the server at a rate of 10Hz, while the server shares global map information with the UAVs at a rate of 3Hz. As shown in Table 4, each UAV sends an average of 450kB of content per second, while the server requires a rate of only 2kB. This shows that the transmission capacity of mainstream routers is more than sufficient to meet the communication requirements of the system, thus validating the feasibility of the system proposed in this paper.

TABLE II. NETWORK TRAFFIC AND TOTAL TIME CONSUMED FOR COMMUNICATION

Sequence	Agent to Server	Server to Agent	Comm Time
MH01	425.54kB/s	2.29kB/s	936.26ms
MH02	435.37kB/s	2.31kB/s	852.33ms
MH03	489.40kB/s	2.31kB/s	807.96ms
Average	450.10kB/s	2.30kB/s	865.51ms

IV. CONCLUSION

We propose a multi-UAV collaborative semantic SLAM system based on a lightweight YOLO model to accomplish the tasks of synchronised localisation and mapping in multi-UAV collaborations. First, to address the issue of limited computational capability of UAVs, we extend the lightweight YOLO model to acquire semantic information in real time. Then, by integrating object detection and multi-view geometric relations, we judge and eliminate the feature points of moving objects, enabling the system to avoid disturbances from moving objects in dynamic environments. Finally, we conducted experimental evaluations using the KITTI, TUM and EuRoC datasets. The experimental results show that our system exhibits commendable performance in feature extraction and trajectory estimation, and meets the requirements for real-time communication, fully demonstrating the feasibility and effectiveness of the proposed system.

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